

15	A small mass undergoes simple harmonic motion about a point O with amplitude x_0 and period T . Its displacement x from O at time $\frac{1}{6}T$ after passing through O is	
	A	$\frac{x_0}{6}$
	B	$\frac{x_0}{2\sqrt{3}}$
	C	$\frac{x_0}{2}$
	D	$\frac{\sqrt{3}x_0}{2}$
Answer: D Since time is measured starting from the instant when the object is at O, we use the sine-function. Using $x = x_0 \sin(\omega t)$ $x = x_0 \sin\left(\frac{2\pi}{T}t\right)$ When $t = \frac{T}{6}$, $x = x_0 \sin\left(\frac{2\pi}{T} \times \frac{T}{6}\right)$ $= x_0 \sin\left(\frac{\pi}{3}\right)$ $= \frac{\sqrt{3}x_0}{2}$		